

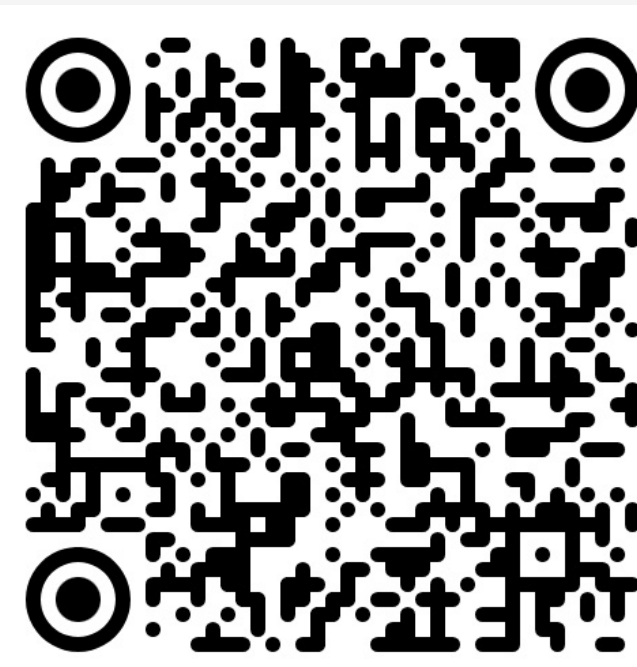
Interactive Strategy Building for Net Zero: A Visual Analytics Approach to Local Decarbonisation

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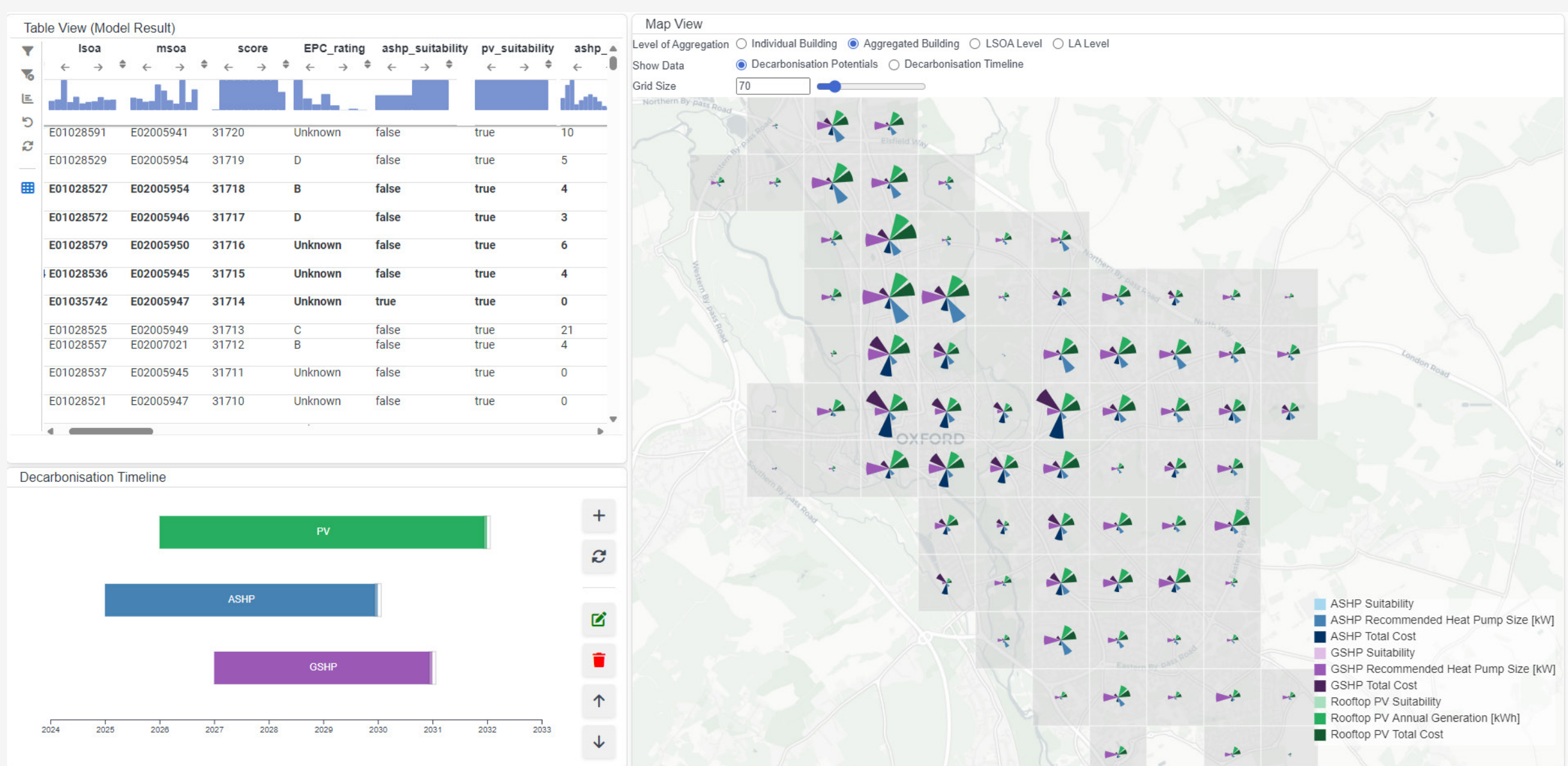
¹City St. George's, University of London



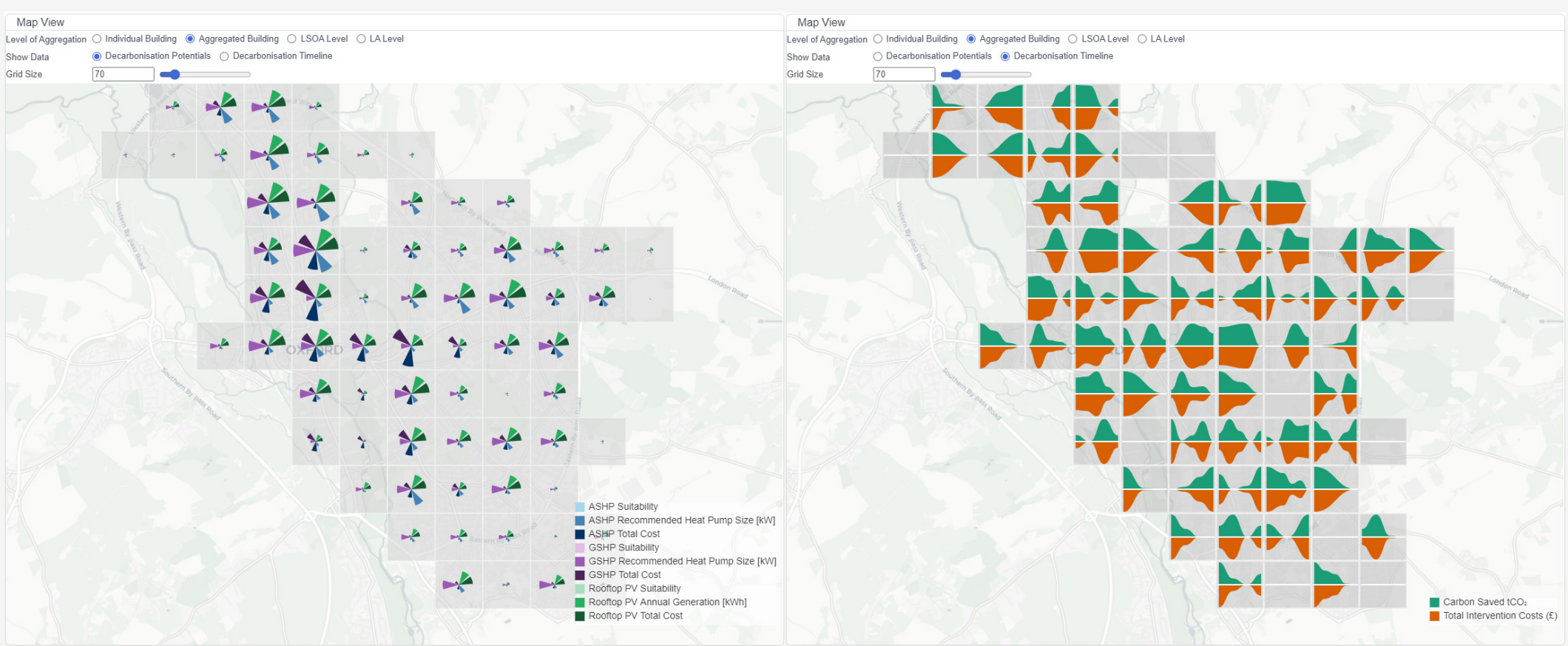
<http://decarb-vis.netlify.app/>

The Challenge

Developing effective and equitable decarbonisation plans is a critical challenge for UK local authorities. Planners must balance technical, economic, and social factors, but current tools often rely on **monolithic, black-box optimisation models** shown as map layers, which can obscure critical trade-offs.



The Decarbonisation Planner Dashboard with linked interactive components: Table View, Timeline View and Map View, showing multivariate glyphs.



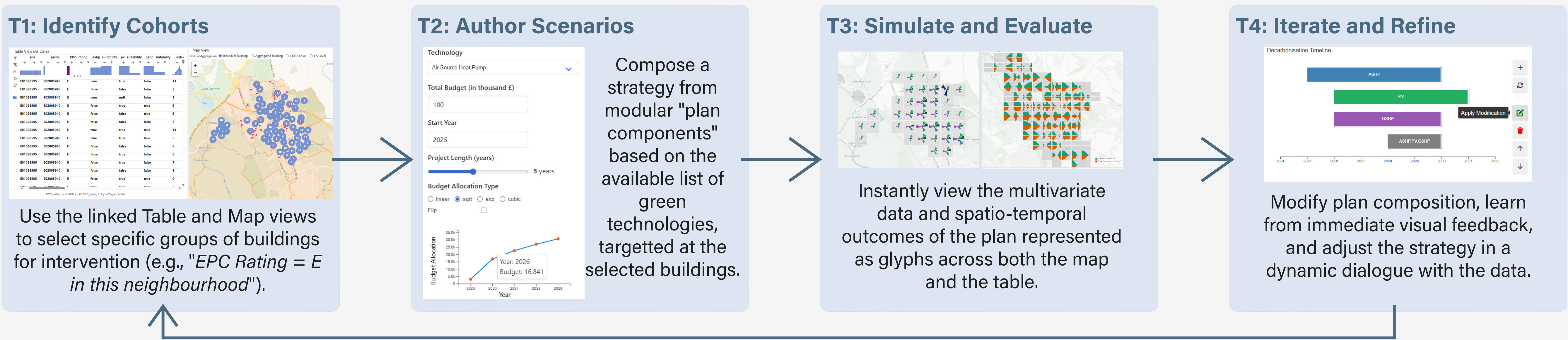
The glyph designs replace overlapping map layers, visualising multivariate and spatio-temporal decarbonisation data and output.

Our Methodology

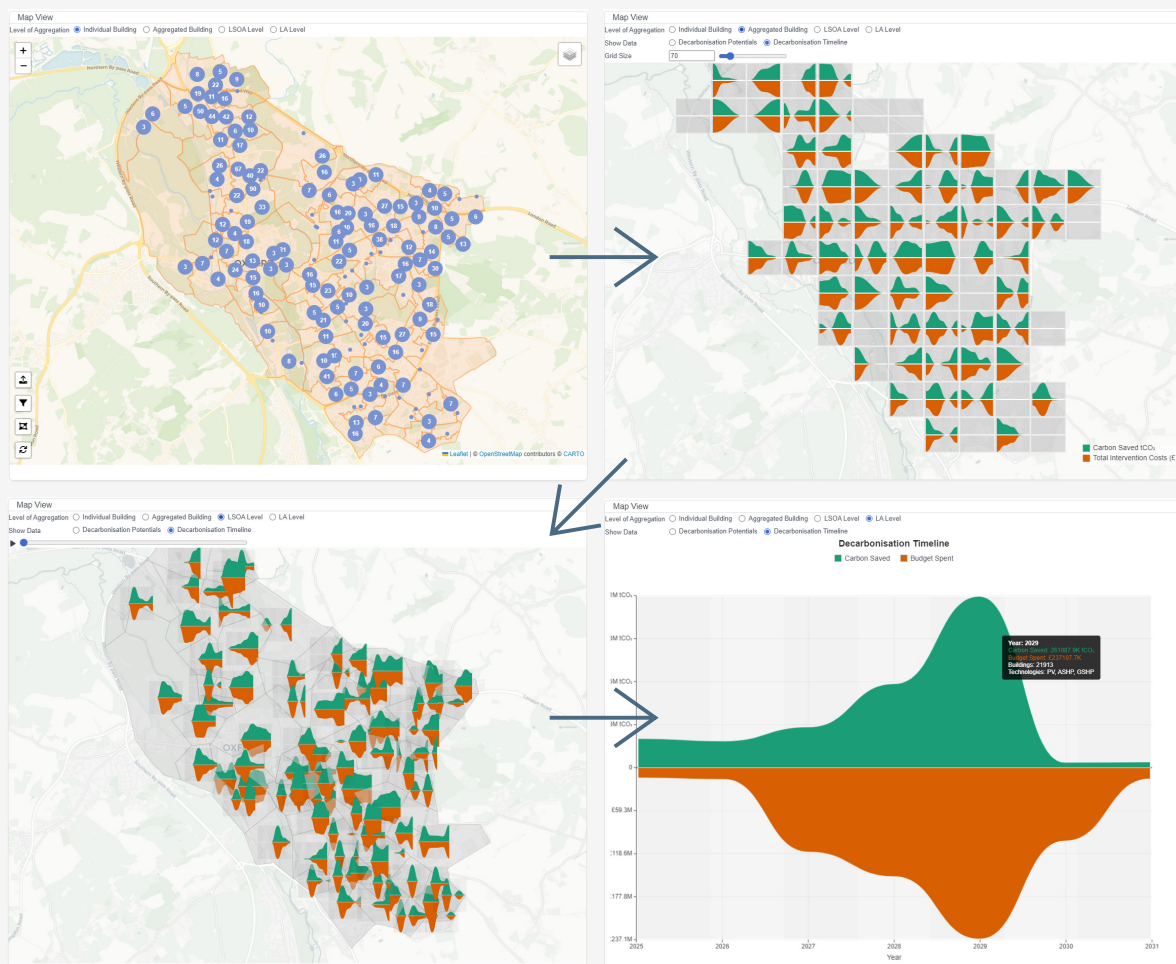
Our design was shaped by collaboration with industry partners through semi-structured dialogues and formative evaluations. We present a human-in-the-loop system with a **modular planning paradigm**, enabling planners to build, simulate, and refine strategies from transparent "plan components" in an iterative workflow. A **multi-scale visual framework** employs model-driven glyphs to represent multivariate and temporal data consistently across geographic scales, supporting fair assessment without cognitive overload.

An Iterative, Data-Driven Workflow

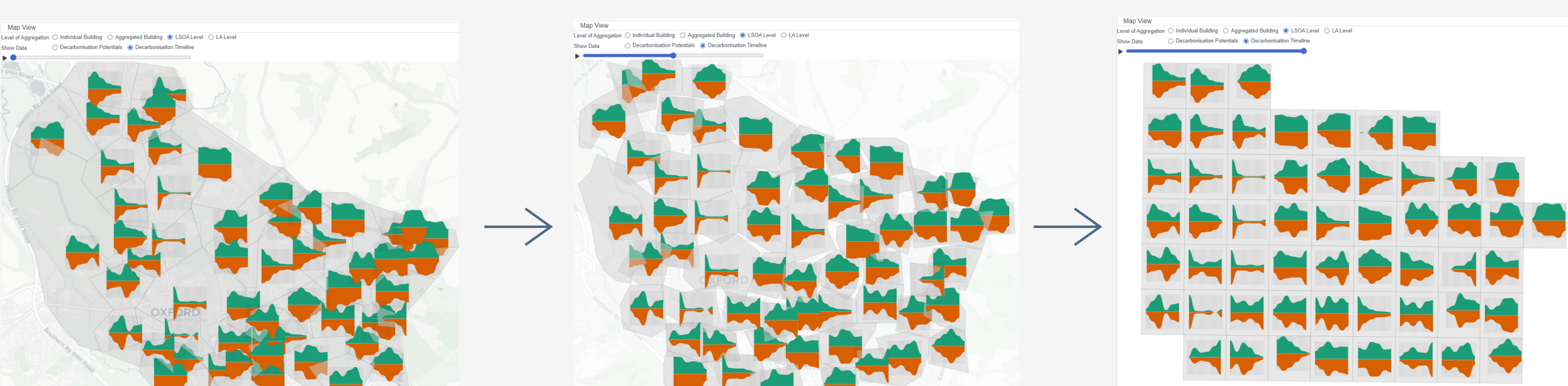
The system engages planners in developing modular decarbonisation scenarios through a continuous cycle of inquiry and refinement:



Visual Design for Complex Trade-offs



Multivariate decarbonisation data across scales, from individual buildings, to aggregated data, to LSOA and finally at LA level, are effectively shown with glyphs, capturing the plan's spatio-temporal properties.



To ensure readability across different levels of aggregation, the system can transform geographies into cartograms, giving each spatial unit (e.g., LSOA) equal visual prominence.

Takeaway and Limitations

- Modular planning makes strategies transparent.
- Glyphs provide at-a-glance insights into trade-offs.
- A human-in-the-loop approach empowers equitable decisions.

Limitations: The model is a strategic simplification; visuals are subject to MAUP; and a formal user study is the next step.

Acknowledgement

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Future work & Collaboration

- Conduct formal user studies with local authorities and industry partner.
- Enhance the simulation model's fidelity.
- Develop a side-by-side scenario comparison tool.
- Incorporate LLMs in the decarbonisation plan creations.



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